

# Integral sums of reciprocals of distinct squarefree semiprimes:

## forty-seven terms are necessary, and the extremal representations of 1

### Abstract

Let  $\mathcal{P} = \{pq : p < q \text{ primes}\}$  be the set of squarefree semiprimes. We prove that if  $T \subset \mathcal{P}$  is a nonempty finite set such that  $\sum_{n \in T} 1/n$  is an integer, then  $|T| \geq 47$ . Since 47-term representations of 1 are known, 47 is the exact minimum; this confirms a conjecture of Watanabe. We also classify the extremal case: there are exactly 23 sets  $T \subset \mathcal{P}$  with  $|T| = 47$  and  $\sum_{n \in T} 1/n = 1$ . Seventeen of them involve only primes at most 73, and each of the other six involves exactly one prime larger than 73, namely 269, 311, 421, 433, 503 or 3779. The proofs are computer-assisted. A  $p$ -adic argument turns integrality into a congruence condition at every vertex of a graph on the primes. A decomposition into a “core” of small primes and a “large-prime part”, together with value, parity and star-size lemmas and an explicit divisor equation for a single edge between two large primes, reduces the problem to a finite exhaustive computation, even though arbitrarily large prime factors are allowed. The computation uses exact integer arithmetic and directed rounding only. We describe independent checks: a second implementation of the completion step, a brute-force validation of the core enumeration on reduced instances, and an exact verification of all 23 solutions. The analytic reduction and the 23 solutions are verified in Lean 4 with Mathlib; there the completeness statements are derived from explicitly stated hypotheses that encode the outcome of the exhaustive searches, and these hypotheses themselves are not verified in Lean.

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# 1 Introduction

An *Egyptian fraction* is a finite sum of distinct unit fractions. Many classical questions ask which rationals are sums of reciprocals of distinct integers from a prescribed set; see [4–6] and [7, Section D11]. This paper studies the set

$$\mathcal{P} = \{pq : p < q \text{ primes}\} = \{6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, \dots\}$$

of *squarefree semiprimes*, i.e. the integers with exactly two distinct prime factors, each to the first power. For a finite set  $T \subset \mathcal{P}$  write

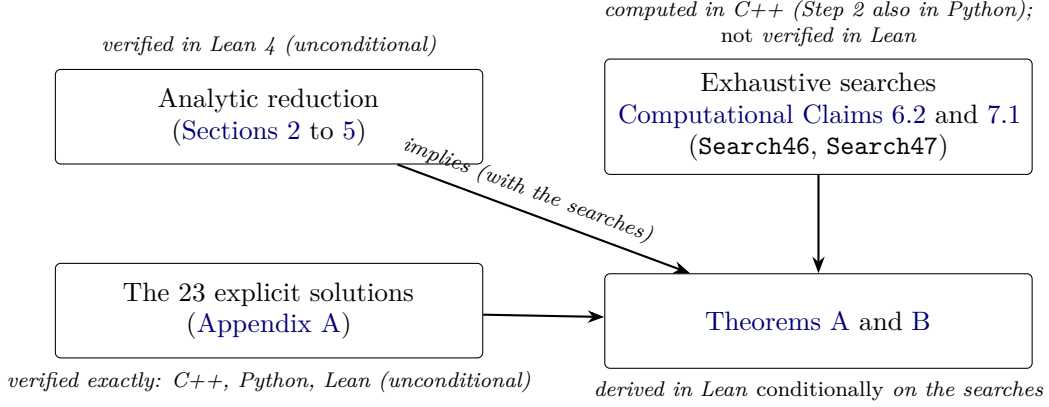
$$\Sigma(T) = \sum_{n \in T} \frac{1}{n}.$$

**Background.** According to the abstract of Watanabe’s preprint [14], in 1978 Johnson found a representation of 1 as a sum of reciprocals of 48 distinct squarefree semiprimes [8]. No shorter representation was known until Watanabe [14] found 17 representations with 47 terms. Watanabe found no representation with fewer than 47 terms and conjectured that 47 is the minimum. His examples were obtained by computer searches in which the prime factors were bounded.<sup>1</sup> A search with bounded prime factors cannot establish minimality, because arbitrarily large primes are allowed in principle; Section 5 shows that primes as large as 3779 really do occur.

In a different direction, Butler, Erdős and Graham [1] proved that every positive integer is a sum of reciprocals of distinct integers that each have exactly three distinct prime factors, and they conjectured the corresponding statement for two prime factors. A recent preprint of Li [9] announces a proof of that conjecture. Egyptian fractions whose denominators are supported on a fixed finite set of primes are studied computationally in [2]. None of these works addresses

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<sup>1</sup>The search space described in [14] has dimension  $325 = \binom{26}{2}$ , the number of squarefree semiprimes whose prime factors are both at most 101. We could not consult the full text while preparing this manuscript, so this detail should be checked against [14].



**Figure 1:** Logical structure of the proof and the verification status of each component.

the minimal number of terms for the set  $\mathcal{P}$ . To the best of our knowledge, no proof that 47 is minimal, and no complete list of the 47-term representations of 1 in which every prime is allowed, has appeared before.

## Results.

**Theorem A** (Minimum number of terms). *Let  $T \subset \mathcal{P}$  be a nonempty finite set such that  $\Sigma(T) \in \mathbb{Z}$ . Then  $|T| \geq 47$ . The bound is sharp: there are sets  $T \subset \mathcal{P}$  with  $|T| = 47$  and  $\Sigma(T) = 1$ .*

**Theorem B** (The extremal representations). *There are exactly 23 sets  $T \subset \mathcal{P}$  with  $|T| = 47$  and  $\Sigma(T) = 1$ . They are listed in [Appendix A](#). Equivalently, there are exactly 23 sets  $T \subset \mathcal{P}$  with  $|T| = 47$  and  $\Sigma(T) \in \mathbb{Z}$ . Of these 23 sets:*

- (i) 17 involve only primes at most 73 (in fact at most 71);
- (ii) each of the other 6 involves exactly one prime  $P > 73$ , where  $P$  is one of 269, 311, 421, 433, 503 and 3779. This prime occurs in exactly three elements  $aP, bP, cP$ , where  $a, b, c \leq 73$  are primes with  $P \mid ab + ac + bc$  (see [Table 2](#)).

In particular, the 17 solutions of (i) are exactly the solutions all of whose prime factors are at most 101. Our 17 solutions of type (i) have the same count as Watanabe’s list of 17 examples. We have not compared the two lists term by term.

**Nature of the proof.** Both theorems are *computer-assisted*. [Sections 2 to 5](#) give a complete mathematical reduction of both statements to finite computations. We state these computations precisely as [Computational Claims 6.2 and 7.1](#). The computations were carried out by the C++ programs `semiprime_reciprocals.cpp` and `semiprime47.cpp`, which use exact integer arithmetic and directed rounding only ([Sections 6, 7 and 10](#)). [Section 8](#) describes the independent checks, and [Section 9](#) describes a Lean 4 formalization of the reduction. [Figure 1](#) summarizes the logical status of each ingredient.

**Why large primes are the main difficulty.** Only finitely many elements of  $\mathcal{P}$  have both prime factors in a given finite set. Since  $\mathcal{P}$  contains  $2P$  for every prime  $P$ , however, no a priori bound on the prime factors of a solution is evident. The main theoretical step is to show that, once the “small-prime” part of a solution is fixed, the large primes are confined to an explicit finite set. A large prime all of whose neighbours are small must divide an explicit integer  $n(A)$

(Lemma 5.1). A single edge joining two large primes leads to a factorization identity with finitely many solutions (Proposition 5.5). The solution with the prime  $3779 = 29 \cdot 37 + 29 \cdot 41 + 37 \cdot 41$  shows that such large primes really occur.

**Organization.** Section 2 gives the size bounds, and Section 3 the graph-theoretic form of integrality. Sections 4 and 5 develop the finite reduction. Sections 6 and 7 describe the two computations and prove Theorems A and B from them. Sections 8 to 10 cover verification, formalization and reproducibility. Section 11 discusses limitations, and Section 12 discloses the use of generative AI.

## 2 Preliminaries and size bounds

Let  $a_1 < a_2 < a_3 < \dots$  be the elements of  $\mathcal{P}$  in increasing order, and let

$$H_k = \sum_{i=1}^k \frac{1}{a_i} \quad (k \geq 0).$$

The first 47 elements of  $\mathcal{P}$  are 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 74, 77, 82, 85, 86, 87, 91, 93, 94, 95, 106, 111, 115, 118, 119, 122, 123, 129, 133, 134, 141, 142, 143, 145, 146, 155, 158.

**Lemma 2.1** (Exchange lemma). *Let  $\mathcal{Q} = \{c_1 < c_2 < \dots\} \subset \mathbb{Z}_{>0}$  be infinite. If  $T \subset \mathcal{Q}$  is finite with  $|T| \leq k$ , then  $\Sigma(T) \leq \sum_{i=1}^k 1/c_i$ . In particular  $\Sigma(T) \leq H_k$  for every  $T \subset \mathcal{P}$  with  $|T| \leq k$ .*

*Proof.* Write  $T = \{t_1 < \dots < t_m\}$  with  $m \leq k$ . The elements  $t_1, \dots, t_i$  are  $i$  distinct elements of  $\mathcal{Q}$ , so  $t_i \geq c_i$ . Hence  $\Sigma(T) = \sum_{i \leq m} 1/t_i \leq \sum_{i \leq m} 1/c_i \leq \sum_{i \leq k} 1/c_i$ .  $\square$

**Proposition 2.2** (Exact values). *The values of  $H_k$  in Table 1 are exact reduced fractions. In particular*

$$H_{35} < 1 - \frac{1757}{79000}, \quad H_{37} < 1 < H_{38}, \quad H_{46} < H_{47} < 2.$$

**Table 1:** Exact values of  $H_k$ . Each inequality in the last column can be read off by comparing numerator and denominator. The decimal expansions (truncated) are for orientation only.

|          | $a_k$ | numerator                  | denominator                | decimal                                |
|----------|-------|----------------------------|----------------------------|----------------------------------------|
| $H_{35}$ | 119   | 14455695312142253588       | 14790464231955481839       | 0.977365... $< 1 - \frac{1757}{79000}$ |
| $H_{37}$ | 123   | 179305550886858336521      | 1804436636298568784358     | 0.993692... $< 1$                      |
| $H_{38}$ | 129   | 1807043389770122629423     | 1804436636298568784358     | 1.001444... $> 1$                      |
| $H_{46}$ | 155   | 94992310862397653755543    | 89805923752009843120899    | 1.057751... $< 2$                      |
| $H_{47}$ | 158   | 15098591040010839136496693 | 14189335952817555213102042 | 1.064080... $< 2$                      |

*Proof.* This is a finite exact computation with rational numbers. It was carried out independently in three ways: with the big-integer routines of both C++ programs (Section 10), in Python with `Fractions.Fraction` (script `paper/verification/verify_paper_facts.py`), and in Lean 4, where the inequalities are proved by `norm_num` (`H35_val`, `H37_lt_one`, `H38_gt_one`, `H46_lt_two`, `H47_lt_two`; Section 9). Lean also proves that  $a_1, \dots, a_{47}$  are the 47 smallest elements of  $\mathcal{P}$ .  $\square$

**Corollary 2.3.** *Let  $T \subset \mathcal{P}$  be nonempty with  $|T| \leq 47$  and  $\Sigma(T) \in \mathbb{Z}$ . Then  $\Sigma(T) = 1$  and  $|T| \geq 38$ .*

*Proof.* By Lemma 2.1 and Proposition 2.2,  $0 < \Sigma(T) \leq H_{47} < 2$ , so  $\Sigma(T) = 1$ . If  $|T| \leq 37$  then  $\Sigma(T) \leq H_{37} < 1$ , a contradiction.  $\square$

Consequently, Theorem A is equivalent to the statement that no set of at most 46 elements of  $\mathcal{P}$  has reciprocal sum 1. Likewise, the 47-element sets with integral sum in Theorem B are exactly those with sum 1.

### 3 The graph of a set of semiprimes and the local criterion

**Definition 3.1.** For a finite  $T \subset \mathcal{P}$  let  $\Gamma(T)$  be the simple graph whose vertices are the primes dividing some element of  $T$  and whose edges are the pairs  $\{p, q\}$  with  $pq \in T$ . For a prime  $p$  let  $N(p) = N_T(p) = \{q : pq \in T\}$  be its set of neighbours ( $N(p) = \emptyset$  if  $p$  is not a vertex).

The map  $T \mapsto \Gamma(T)$  is a bijection between finite subsets of  $\mathcal{P}$  and finite simple graphs on the set of primes without isolated vertices. The number of edges is  $|T|$ . For a prime  $p$  and an integer  $m$  with  $p \nmid m$  we write  $m^{-1}$  for the inverse of  $m$  modulo  $p$ .

**Lemma 3.2** (Local criterion). *For every finite  $T \subset \mathcal{P}$ ,*

$$\Sigma(T) \in \mathbb{Z} \iff \sum_{q \in N(p)} q^{-1} \equiv 0 \pmod{p} \text{ for every prime } p. \quad (1)$$

*Proof.* Let  $\nu_p$  denote the  $p$ -adic valuation on  $\mathbb{Q}$ . A rational number is an integer if and only if its  $p$ -adic valuation is nonnegative for every prime  $p$ . Fix  $p$  and split

$$\Sigma(T) = \frac{1}{p} A_p + B_p, \quad A_p = \sum_{q \in N(p)} \frac{1}{q}, \quad B_p = \sum_{n \in T, p \nmid n} \frac{1}{n}.$$

Every element of  $T$  is squarefree, so every denominator in  $B_p$  and in  $A_p$  is prime to  $p$ , and  $\nu_p(B_p) \geq 0$ . Write  $A_p = u/v$  with  $v = \prod_{q \in N(p)} q$  and  $u = \sum_{q \in N(p)} v/q$ . Then  $p \nmid v$  and  $u \equiv v \sum_{q \in N(p)} q^{-1} \pmod{p}$ . If the congruence in (1) holds at  $p$ , then  $p \mid u$ , so  $\nu_p(A_p/p) \geq 0$  and hence  $\nu_p(\Sigma(T)) \geq 0$ . Conversely, if  $\nu_p(\Sigma(T)) \geq 0$ , then  $\nu_p(A_p/p) = \nu_p(\Sigma(T) - B_p) \geq 0$ , so  $p \mid u$  and the congruence holds. For a prime  $p$  that divides no element of  $T$  both sides are trivially true.  $\square$

**Corollary 3.3.** *If  $\Sigma(T) \in \mathbb{Z}$ , then no vertex of  $\Gamma(T)$  has degree 1.*

*Proof.* If  $N(p) = \{q\}$  then  $q^{-1} \not\equiv 0 \pmod{p}$ .  $\square$

For every element of the 23 solutions of Theorem B, the congruences (1) were checked directly at every vertex, as an independent test of the implementation. In Lean the lemma is `integral_iff_localCond`. Its proof uses the common denominator  $\prod_{p \in \Gamma(T)} p$  instead of valuations.

### 4 Core decomposition and the finite reduction

Throughout this section  $K \in \{46, 47\}$  is a *budget*, and  $T \subset \mathcal{P}$  is a finite set with

$$\Sigma(T) = 1, \quad |T| \leq K. \quad (2)$$

By Lemma 3.2, the congruence (1) holds at every prime.

**Definition 4.1** (Small and large primes). Let  $\mathcal{S} = \{2, 3, 5, \dots, 73\}$  be the set of the 21 primes at most 73. Call the primes  $P > 73$  *large*; the smallest large primes are  $b_0 = 79 < b_1 = 83 < b_2 = 89 < \dots$ . Let  $\mathcal{P}_{\mathcal{S}} = \{pq \in \mathcal{P} : p, q \in \mathcal{S}\}$ , a set of  $\binom{21}{2} = 210$  elements.

The primes dividing  $a_1, \dots, a_{46}$  are exactly the elements of  $\mathcal{S}$  (since  $146 = 2 \cdot 73 \leq a_{46} = 155 < a_{47} = 158 = 2 \cdot 79$ ). The reduction below is valid for any finite cutoff; the choice of  $\mathcal{S}$  only affects the efficiency of the computation.

**Definition 4.2** (Core, large part, residues). For  $T$  as in (2) define:

- the *core*  $C = T \cap \mathcal{P}_{\mathcal{S}}$  and the *large part*  $G = T \setminus C$ ;
- for  $q \in \mathcal{S}$  the *core residue*  $\rho(q) = \sum_{r: qr \in C} r^{-1} \in \mathbb{Z}/q\mathbb{Z}$ ;
- the *unsatisfied set*  $U = U(C) = \{q \in \mathcal{S} : \rho(q) \neq 0\}$ ;
- for  $q \in \mathcal{S}$  the number  $d_q$  of large neighbours of  $q$ ;
- the number  $b$  of *large-large* elements of  $G$ , i.e. elements with both prime factors large.

Every element of  $G$  is either  $qP$  with  $q \in \mathcal{S}$  and  $P$  large, or  $PP'$  with  $P, P'$  both large. In particular every element of  $G$  has at most one prime factor in  $\mathcal{S}$ . The quantities  $\rho$  and  $U$  depend only on  $C$ .

**Lemma 4.3** (Unsatisfied primes). *Every  $q \in U$  has a large neighbour. Hence  $|G| \geq |U|$  and*

$$|C| + |U| \leq |T| \leq K.$$

*Proof.* At  $q \in \mathcal{S}$  the congruence (1) reads  $\rho(q) + \sum_P P^{-1} \equiv 0 \pmod{q}$ , where  $P$  runs over the large neighbours of  $q$ . If  $\rho(q) \neq 0$ , this sum is nonempty. Choosing one element  $e(q) = qP \in G$  for each  $q \in U$  gives an injection  $U \rightarrow G$ , because an element of  $G$  has at most one prime factor in  $\mathcal{S}$ .  $\square$

For  $x \geq 0$  let  $V(x)$  be the sum of the  $x$  largest values of  $1/n$  over the elements  $n \in \mathcal{P}$  that have a large prime factor. Thus  $V(x) = \frac{1}{158} + \frac{1}{166} + \frac{1}{178} + \frac{1}{194} + \frac{1}{202} + \frac{1}{206} + \dots$  ( $x$  terms). Since every such  $n$  is at least  $2 \cdot 79 = 158$ , we have  $V(x) \leq x/158$ .

**Lemma 4.4** (Value bound, general form). *For  $T$  as in (2) we have  $\Sigma(C) \leq 1$ . Moreover, for every subset  $U' \subseteq U$ ,*

$$1 \leq \Sigma(C) + \sum_{q \in U'} \frac{1}{79q} + V(K - |C| - |U'|). \quad (3)$$

*Proof.* First,  $\Sigma(C) = 1 - \Sigma(G) \leq 1$ . Take the elements  $e(q)$ ,  $q \in U'$ , from the proof of Lemma 4.3. They are distinct, and  $e(q) = qP \geq 79q$ . The remaining  $|G| - |U'| \leq K - |C| - |U'|$  elements of  $G$  are distinct elements of  $\mathcal{P}$  with a large prime factor. By Lemma 2.1 their reciprocal sum is at most  $V(K - |C| - |U'|)$ . Now use  $1 = \Sigma(C) + \Sigma(G)$ .  $\square$

**Definition 4.5** (Admissible cores). A set  $C \subseteq \mathcal{P}_{\mathcal{S}}$  is *K-admissible* if  $|C| + |U(C)| \leq K$  and  $\Sigma(C) \leq 1$ , and if (3) holds for every upper segment  $U' = U(C) \cap (p, \infty)$ ,  $p \geq 0$ . (The upper segments include  $U(C)$  itself, for  $p = 0$ , and  $\emptyset$ , for  $p \geq 73$ .)

By Lemmas 4.3 and 4.4, the core of every  $T$  satisfying (2) is *K-admissible*.

*Remark 4.6* (Why upper segments). The search of [Section 6](#) processes the primes of  $\mathcal{S}$  in decreasing order, and at an intermediate stage it uses (3) with  $U'$  equal to the set of primes *already known* to lie in  $U$ . That set is an upper segment of  $U$ . A prime that joins  $U$  later, for instance  $q = 2$  at the last level, adds  $1/(79q)$  to the right-hand side of (3) but removes one term of  $V$ , possibly a much smaller one. So the right-hand side for  $U' = U$  can be larger than for a proper upper segment. Thus the set of cores produced by the search is *not* the set of cores satisfying (3) for  $U' = U$  alone, as stated in an earlier description of the method. It contains the set of admissible cores in the sense of [Definition 4.5](#), and possibly some further cores, because the bounds used at intermediate nodes are not tight. Pruning with upper segments is sound because (3) holds for every  $U' \subseteq U$ . On reduced instances the difference is substantial ([Section 8.4](#)).

**Definition 4.7** (Shapes). Let  $C$  be a core with residues  $\rho$  and unsatisfied set  $U$ . For  $q \in U$  let  $P_{\min}(q)$  be the least large prime  $P$  with  $P^{-1} \equiv -\rho(q) \pmod{q}$ . Put

$$w_q(0) = 0, \quad w_q(1) = \frac{1}{q P_{\min}(q)}, \quad w_q(d) = \sum_{k=0}^{d-1} \frac{1}{q b_k} \quad (d \geq 2).$$

A pair  $(d, b)$ , with  $d = (d_q)_{q \in \mathcal{S}}$  and  $b$  nonnegative integers, is a  $K$ -feasible shape for  $C$  if

(S1)  $d_q \geq 1$  for  $q \in U$  and  $d_q \neq 1$  for  $q \in \mathcal{S} \setminus U$ ;

(S2)  $E := b + \sum_{q \in U} (d_q - 1) + \sum_{q \in \mathcal{S} \setminus U} d_q \leq K - |C| - |U|$ ;

(S3)  $\rho(2) + d_2 \equiv 0 \pmod{2}$ , where  $\rho(2) \in \{0, 1\}$ ;

(S4)  $1 - \Sigma(C) \leq \sum_{q \in \mathcal{S}} w_q(d_q) + \frac{b}{79 \cdot 83}$ .

Dirichlet's theorem guarantees that  $P_{\min}(q)$  exists, but we do not need it. If no large prime lies in the required class, then  $d_q = 1$  is impossible by [Lemma 4.8](#) below. The programs compute  $P_{\min}(q)$  by direct search, and this search terminated for every class that occurred. The Lean formalization defines  $P_{\min}(q) = 0$  if no such prime exists.

**Lemma 4.8** (Shape lemma). *For  $T$  as in (2), the pair  $(d, b)$  of [Definition 4.2](#) is a  $K$ -feasible shape for the core  $C$  of  $T$ .*

*Proof.* (S1): For  $q \in U$  use [Lemma 4.3](#). If  $q \notin U$  has exactly one large neighbour  $P$ , then (1) at  $q$  gives  $P^{-1} \equiv 0 \pmod{q}$ , which is impossible.

(S2): Each element of  $G$  has exactly one small prime factor or none, so  $|G| = \sum_{q \in \mathcal{S}} d_q + b$ . Hence  $E = |G| - |U| \leq K - |C| - |U|$ .

(S3): Every large prime  $P$  is odd, so  $P^{-1} \equiv 1 \pmod{2}$ . The congruence (1) at 2 becomes  $\rho(2) + d_2 \equiv 0 \pmod{2}$ .

(S4): Split

$$1 - \Sigma(C) = \Sigma(G) = \sum_{q \in \mathcal{S}} \sum_{P \text{ large}, qP \in T} \frac{1}{qP} + \sum_{PP' \in G \text{ large-large}} \frac{1}{PP'}.$$

If  $d_q \geq 2$ , the  $d_q$  large neighbours of  $q$  are distinct primes, so their contribution is at most  $w_q(d_q)$  by [Lemma 2.1](#). If  $d_q = 1$ , then  $q \in U$  by (S1). By (1) the unique large neighbour  $P$  satisfies  $P^{-1} \equiv -\rho(q)$ , so  $P \geq P_{\min}(q)$  and  $1/(qP) \leq w_q(1)$ . A large-large element is  $PP'$  with  $79 \leq P < P'$ , so  $P' \geq 83$  and  $1/(PP') \leq 1/(79 \cdot 83)$ .  $\square$

**The finite reduction.** [Lemmas 4.3](#), [4.4](#) and [4.8](#) reduce the problem as follows.

(1) The core is one of finitely many admissible subsets of the 210 elements of  $\mathcal{P}_{\mathcal{S}}$ .

- (2) The shape is one of finitely many, because all  $d_q$  and  $b$  are bounded by (S2).
- (3) The large part is then subject to the constraints of Section 5: a large prime whose neighbours are all small divides an explicit positive integer, and a single large-large edge is governed by a factorization identity. Both confine the large primes to explicit finite sets.

The only case not reduced by theory is  $b \geq 2$ . The computation shows that no feasible shape with  $b \geq 2$  exists for  $K = 47$ , and none with  $b \geq 1$  for  $K = 46$  (Sections 6 and 7). If such shapes had survived, further analysis would have been needed; the programs were written to report them as “unresolved” rather than discard them.

## 5 Large primes: stars and the large-large component

### 5.1 Stars

For a finite set  $A$  of primes put

$$n(A) = \sum_{q \in A} \prod_{r \in A \setminus \{q\}} r,$$

so that  $\sum_{q \in A} 1/q = n(A) / \prod_{r \in A} r$ . For instance  $n(\{a, b\}) = a + b$  and  $n(\{a, b, c\}) = ab + ac + bc$ .

**Lemma 5.1** (Star lemma). *Let  $P$  be a prime with  $P \notin A$ . Then  $\sum_{q \in A} q^{-1} \equiv 0 \pmod{P}$  if and only if  $P \mid n(A)$ . Consequently, if  $\Sigma(T) \in \mathbb{Z}$  and a large prime  $P$  has all its neighbours in  $\mathcal{S}$ , then  $P \mid n(N(P))$  and in particular  $P \leq n(N(P))$ . Moreover  $|N(P)| \geq 2$ .*

*Proof.* Multiply the congruence by  $\prod_{r \in A} r$ , which is invertible modulo  $P$ . The second statement follows from (1) at  $P$  and  $n(A) > 0$ ; the third follows from Corollary 3.3.  $\square$

We call such a pair  $(P, A)$  with  $A = N(P) \subseteq \mathcal{S}$  a *star* with *centre*  $P$  and *leaves*  $A$ . For a fixed set  $A$  of leaves only the finitely many prime divisors of  $n(A)$  can be centres. This is the mechanism that bounds the large primes.

**Lemma 5.2** (Star size). *Suppose  $T$  satisfies (2) and  $P$  is a large prime all of whose neighbours lie in  $\mathcal{S}$ . Then  $|N(P)| \leq |T| - 36 \leq K - 36$ . Thus a star has at most 10 leaves if  $K = 46$  and at most 11 if  $K = 47$ .*

*Proof.* Let  $d = |N(P)|$ . The  $d$  elements  $qP$  contribute at most  $\frac{1}{79} \sum_{q \in \mathcal{S}} \frac{1}{q} \leq \frac{1757}{79000}$ , since  $\sum_{q \in \mathcal{S}} 1/q = 1.7565 \dots \leq 1.757$  (exact computation). If  $d \geq |T| - 35$ , the other at most 35 elements contribute at most  $H_{35} < 1 - \frac{1757}{79000}$  by Lemma 2.1 and Proposition 2.2. Then  $\Sigma(T) < 1$ , a contradiction.  $\square$

The programs use the slightly sharper bound  $\frac{1}{79} \sum_{i < d} 1/p_i + H_{K-d} \geq 1$ , with  $p_0 < p_1 < \dots$  the primes of  $\mathcal{S}$ . Recomputed with exact rationals, it gives the same maxima, 10 and 11.

**Definition 5.3** (Candidate stars). Let  $C$  be a core and  $(d, b)$  a feasible shape, and put  $W = \{q \in \mathcal{S} : d_q \geq 1\}$ . A *candidate star* for  $(C, d)$  with at most  $m$  leaves is a pair  $(P, A)$  with  $A \subseteq W$ ,  $2 \leq |A| \leq m$ ,  $P > 73$  prime,  $P \mid n(A)$ , and  $P^{-1} \equiv -\rho(q) \pmod{q}$  for every  $q \in A$  with  $d_q = 1$ .

Since  $P \leq n(A) \leq |A| \prod_{r \in A} r \leq 21 \prod_{r \in \mathcal{S}} r$ , there are only finitely many candidate stars (starCand\_finite). If  $b = 0$ , every large prime of  $T$  is the centre of a star, and the stars of  $T$  have the following properties:

- their centres are distinct;
- $q \in \mathcal{S}$  is a leaf of exactly  $d_q$  of them;

- each of them is a candidate star with at most  $K - 36$  leaves ([Lemmas 5.1 and 5.2](#)). The residue condition holds because a leaf  $q$  with  $d_q = 1$  has the centre as its only large neighbour;
- their total value  $\sum_{(P,A)} \sum_{q \in A} 1/(qP)$  equals  $1 - \Sigma(C)$ .

Conversely, these conditions determine  $G$ . So for  $b = 0$  the completions of a core are exactly the *exact covers* of the multiplicities  $(d_q)$  by candidate stars with distinct centres and total value  $1 - \Sigma(C)$ .

**The six solutions with a large prime.** [Table 2](#) lists the six solutions of [Theorem B](#) that contain a large prime. In each of them  $G$  consists of a single star with three leaves  $a, b, c$ . The core has 44 elements,  $U = \{a, b, c\}$ ,  $d_a = d_b = d_c = 1$  and all other  $d_q = 0$ , so the excess is  $E = 0$  and  $|C| + |U| = 47$ . The large prime divides  $n(\{a, b, c\}) = ab + ac + bc$ . For  $P = 311$ , 503 and 3779 the quantity  $ab + ac + bc$  is itself prime and equal to  $P$ ; for  $P = 269$ , 421 and 433 it is  $3P$ ,  $11P$  and  $7P$ , respectively.

**Table 2:** The six 47-term representations of 1 that contain a prime  $P > 73$ . The last column refers to the numbering of [Appendix A](#).

| $P$  | leaves $A = \{a, b, c\}$ | $n(A) = ab + ac + bc$ | large elements $aP, bP, cP$ | no. |
|------|--------------------------|-----------------------|-----------------------------|-----|
| 269  | $\{5, 11, 47\}$          | $807 = 3 \cdot 269$   | 1345, 2959, 12643           | 11  |
| 311  | $\{3, 5, 37\}$           | 311                   | 933, 1555, 11507            | 2   |
| 421  | $\{29, 43, 47\}$         | $4631 = 11 \cdot 421$ | 12209, 18103, 19787         | 12  |
| 433  | $\{3, 37, 73\}$          | $3031 = 7 \cdot 433$  | 1299, 16021, 31609          | 1   |
| 503  | $\{3, 13, 29\}$          | 503                   | 1509, 6539, 14587           | 3   |
| 3779 | $\{29, 37, 41\}$         | 3779                  | 109591, 139823, 154939      | 5   |

**Example 5.4** (The prime 3779). Solution 5 of [Appendix A](#) contains  $109591 = 29 \cdot 3779$ ,  $139823 = 37 \cdot 3779$  and  $154939 = 41 \cdot 3779$ , and  $3779 = 29 \cdot 37 + 29 \cdot 41 + 37 \cdot 41$  is prime. Any search that bounds the prime factors by a fixed cutoff below 3779 misses this representation of 1. Such a cutoff includes the natural choices 101 and 1000. The present argument bounds large primes only through divisibility: a star centre divides  $n(A)$  for its set of leaves  $A$ , which for three leaves can be as large as  $67 \cdot 71 + 67 \cdot 73 + 71 \cdot 73 = 14831$ .

## 5.2 One edge between two large primes

Suppose  $b = 1$ . Then  $G$  contains exactly one element  $P_1 P_2$  with  $P_1 \neq P_2$  both large. Every other large prime has only small neighbours, since a second large neighbour would create a second large-large element, and is therefore the centre of a star. For  $i = 1, 2$  we have  $N(P_i) = A_i \cup \{P_{3-i}\}$  with  $A_i \subseteq \mathcal{S}$ , and  $A_i \neq \emptyset$  by [Corollary 3.3](#). Put

$$b_i = \prod_{q \in A_i} q, \quad a_i = n(A_i), \quad Z = \sum_{q \in A_1} \frac{1}{qP_1} + \sum_{q \in A_2} \frac{1}{qP_2} + \frac{1}{P_1 P_2},$$

the value of the component. Finally put

$$W = a_1 b_2 P_2 + a_2 b_1 P_1 + b_1 b_2, \quad t = Z b_1 b_2, \quad N = b_1 b_2 (t + a_1 a_2).$$

**Proposition 5.5** (The component equation). *Let  $P_1 \neq P_2$  be primes not dividing  $b_1 b_2$ , and let  $A_1, A_2$  be nonempty.*

- (a)  $Z b_1 b_2 P_1 P_2 = W$ ; that is,  $t P_1 P_2 = W$  in  $\mathbb{Q}$ .

- (b) The local conditions (1) at  $P_1$  and at  $P_2$  hold if and only if  $P_1P_2 \mid W$ , i.e. if and only if  $t \in \mathbb{Z}$ .
- (c) For every rational  $t \neq 0$ ,

$$(tP_1 - a_1b_2)(tP_2 - a_2b_1) = b_1b_2(t + a_1a_2) \iff tP_1P_2 = W.$$

In particular the identity on the left holds for  $t = Zb_1b_2$ , whether or not  $t$  is an integer.

- (d) Suppose  $t$  is a positive integer with  $tP_1P_2 = W$ , and put  $X = tP_1 - a_1b_2$ ,  $Y = tP_2 - a_2b_1$ . Then  $X$  and  $Y$  are positive integers with  $XY = N$ , and

$$P_1 = \frac{X + a_1b_2}{t}, \quad P_2 = \frac{N/X + a_2b_1}{t}.$$

Conversely, if  $t > 0$  and  $X$  is a positive divisor of  $N$  such that both quotients are integers, then they satisfy  $tP_1P_2 = W$ .

*Proof.* (a) Use  $\sum_{q \in A_i} 1/q = a_i/b_i$  and clear denominators.

(b) Since  $N(P_1) = A_1 \cup \{P_2\}$ , Lemma 5.1 shows that (1) at  $P_1$  is equivalent to  $P_1 \mid n(A_1 \cup \{P_2\}) = a_1P_2 + b_1$ . Now  $W = b_2(a_1P_2 + b_1) + a_2b_1P_1 \equiv b_2(a_1P_2 + b_1) \pmod{P_1}$  and  $P_1 \nmid b_2$ , so this is equivalent to  $P_1 \mid W$ . The same argument applies at  $P_2$ . Since  $P_1 \neq P_2$ , both conditions together are equivalent to  $P_1P_2 \mid W$ , which by (a) says  $t \in \mathbb{Z}$ .

(c) Expanding gives  $(tP_1 - a_1b_2)(tP_2 - a_2b_1) - b_1b_2(t + a_1a_2) = t(tP_1P_2 - W)$ .

(d) From  $tP_1P_2 = W$  we get  $P_2X = W - a_1b_2P_2 = a_2b_1P_1 + b_1b_2 > 0$ , so  $X > 0$ , and similarly  $Y > 0$ . Then (c) gives  $XY = N$ . The converse is (c) read backwards.  $\square$

*Remark 5.6 (Correction).* An earlier description of the method stated that the factorization identity in (c) is itself equivalent to the local conditions at  $P_1$  and  $P_2$ . By (a) and (c), however, the identity holds automatically for  $t = Zb_1b_2$ . What is equivalent to the local conditions is the *integrality* of  $t$ , by (b). The positivity of  $X$  and  $Y$  and the nonemptiness of  $A_1$  and  $A_2$ , both used by the program, are proved above.

**Finite enumeration for  $b = 1$ .** Fix a core  $C$ , a feasible shape with  $b = 1$ , nonempty attachment sets  $A_1, A_2 \subseteq W$ , and a family  $F$  of stars covering the remaining multiplicities  $d_q - [q \in A_1] - [q \in A_2]$ . Then the value of the component is  $Z = 1 - \Sigma(C) - \Sigma(F)$ , and  $t = Zb_1b_2$  is determined; it must be a positive integer. By Proposition 5.5(d) the pair  $(P_1, P_2)$  is then determined by a positive divisor  $X$  of the explicit integer  $N$ . All choices of  $A_1$ ,  $A_2$  and  $F$  range over finite sets, so the  $b = 1$  completions of a core form a finite, explicitly enumerable set (`comp_pairs_finite`). Interchanging  $(A_1, P_1)$  and  $(A_2, P_2)$  replaces  $X$  by  $N/X$ , so unordered pairs  $\{A_1, A_2\}$  suffice.

## 6 Computer-assisted proof of the lower bound

This section describes the computation carried out by `semiprime_reciprocals.cpp` for  $K = 46$  and proves Theorem A from its outcome. The computation consists of two steps: enumeration of all admissible cores (Step 1) and, for each of them, enumeration of all feasible shapes and completions (Step 2).

## 6.1 Step 1: enumeration of the admissible cores

Let  $p_0 = 2 < p_1 < \dots < p_{20} = 73$  be the primes of  $\mathcal{S}$ . Every core  $C \subseteq \mathcal{P}_{\mathcal{S}}$  is determined by the sets

$$L_j = \{p_i : i < j, p_i p_j \in C\} \quad (j = 20, 19, \dots, 1)$$

of *lower neighbours*. The search is a depth-first search that chooses  $L_{20}, L_{19}, \dots, L_1$  in this order (Algorithm 1). When  $L_j$  is chosen, every core edge at  $p_j$  is known, because the edges to larger primes were fixed at earlier levels. Hence  $\rho(p_j)$  is known, and it is decided whether  $p_j \in U$ . At every stage the set of primes already known to lie in  $U$  is therefore an upper segment  $U \cap (p, \infty)$  of the final set  $U$ . Each core corresponds to exactly one leaf of the search tree, so each core is visited at most once.

For each  $j$ , the choice of  $L_j$  is split into two parts. The primes  $p_i$  with  $8 \leq i < j$  are handled by explicit inclusion/exclusion recursion. The 8 smallest primes are handled by a precomputed table of all  $2^8$  subsets, indexed by their contribution to the residue at  $p_j$ . This lets the search jump directly to the subsets that make  $p_j$  satisfied, and to the others (which put  $p_j$  into  $U$ ) only when that is still affordable.

**Fixed-point bounds.** Write  $\text{up}(n) = \lceil 2^{96}/n \rceil$  and  $\text{lo}(n) = \lfloor 2^{96}/n \rfloor$ . For a finite set  $X$  of positive integers put  $\text{up}(X) = \sum_{n \in X} \text{up}(n)$  and  $\text{lo}(X) = \sum_{n \in X} \text{lo}(n)$ . Then

$$\text{lo}(X) \leq 2^{96} \Sigma(X) \leq \text{up}(X).$$

Let  $v_1 < v_2 < \dots$  be the elements of  $\mathcal{P}$  with a large prime factor, and put  $\text{VB}(x) = \sum_{i \leq x} \text{up}(v_i) \geq 2^{96} V(x)$ . All these quantities are integers below  $2^{128}$  and are computed exactly in unsigned 128-bit arithmetic.

**Pruning tests.** Consider a node of the search at which the decided core edges form a set  $C_0$  and the decided unsatisfied primes form  $U_0$ . Let  $j$  be the current level and  $i$  the number of lower primes whose membership in  $L_j$  is still open. The search discards the node if one of the following holds:

$$(P1) \quad |C_0| + |U_0| > K;$$

$$(P2) \quad \text{lo}(C_0) > 2^{96};$$

$$(P3) \quad \text{up}(C_0) + \sum_{q \in U_0} \text{up}(79q) + B_{j,i}(R) < 2^{96}, \text{ where } R = K - |C_0| - |U_0| \text{ and}$$

$$B_{j,i}(R) = \max_{\substack{x_1 + x_2 \leq R \\ x_1 \leq i}} \left( M_{j,i}(x_1) + M'_j(x_2) + \text{VB}(R - x_1 - x_2) \right).$$

Here  $M_{j,i}(x)$  is the sum of the  $x$  largest values  $\text{up}(p_j p_k)$  with  $k < i$  (possible further edges at  $p_j$ ), and  $M'_j(x)$  is the sum of the  $x$  largest values  $\text{up}(p_k p_l)$  with  $k < l < j$  (possible edges among smaller primes).

Test (P3) is also applied to the tentative set  $U_0 \cup \{p_j\}$ , with  $R$  replaced by  $R - 1$ , before  $p_j$  is put into  $U$ . At a leaf, where the complete core  $C$  and  $U$  are known, the core is kept only if (P1) and (P2) hold and

$$\text{up}(C) + \sum_{q \in U} \text{up}(79q) + \text{VB}(K - |C| - |U|) \geq 2^{96}. \quad (4)$$

---

**Algorithm 1** Step 1 (core enumeration), schematically

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```

1: procedure LEVEL( $j, C_0, U_0$ )                                 $\triangleright$  primes  $> p_j$  fully processed
2:   if  $j = 0$  then                                            $\triangleright$  prime 2: no lower neighbours
3:     decide  $2 \in U$  from  $\rho(2)$ ; apply (P1); LEAF( $C_0, U_0$ ); return
4:   for all  $L \subseteq \{p_0, \dots, p_{j-1}\}$  not excluded by (P1)–(P3) do     $\triangleright$  explicit + table
5:      $C_1 \leftarrow C_0 \cup \{p_j q : q \in L\}$ ; compute  $\rho(p_j)$  and the residue updates at  $q \in L$ 
6:      $U_1 \leftarrow U_0 \cup \{p_j\}$  if  $\rho(p_j) \neq 0$  else  $U_0$            $\triangleright$  (P3) tested for  $U_1$ 
7:     LEVEL( $j - 1, C_1, U_1$ )
8: procedure LEAF( $C, U$ )
9:   if (4) holds then output  $C$  to Step 2

```

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**Lemma 6.1** (Soundness of Step 1). *Every  $K$ -admissible core (Definition 4.5) reaches a leaf of the search and is passed to Step 2.*

*Proof.* Let  $C$  be admissible, with unsatisfied set  $U$ , and consider a node on the path to  $C$ , with  $C_0 \subseteq C$  and  $U_0 = U \cap (p, \infty)$  for some  $p$ . (P1) fails because  $|C_0| + |U_0| \leq |C| + |U| \leq K$ . (P2) fails because  $\text{lo}(C_0) \leq 2^{96} \Sigma(C_0) \leq 2^{96} \Sigma(C) \leq 2^{96}$ . For (P3), let  $x_1$  and  $x_2$  be the numbers of edges of  $C \setminus C_0$  at  $p_j$  and among the smaller primes, respectively. Then  $\text{up}(C \setminus C_0) \leq M_{j,i}(x_1) + M'_j(x_2)$ . Also  $R - x_1 - x_2 = K - |C| - |U_0|$ , and  $\text{VB}(x) \geq 2^{96} V(x)$ . Hence

$$\text{up}(C_0) + \sum_{q \in U_0} \text{up}(79q) + B_{j,i}(R) \geq 2^{96} \left( \Sigma(C) + \sum_{q \in U_0} \frac{1}{79q} + V(K - |C| - |U_0|) \right) \geq 2^{96},$$

where the last step is (3) for the upper segment  $U' = U_0$ . The tentative test with  $U_0 \cup \{p_j\}$  is again a test of this form for the upper segment  $U \cap [p_j, \infty)$ , and (4) is the case  $U' = U$ .  $\square$

Because of the rounding, the search may keep a few cores that are not admissible. This is harmless: they are examined further in Step 2.

**Parallelization.** The search tree above a fixed level (the prime 61) is traversed by every thread in the same deterministic order, and the nodes at that level are numbered consecutively. Each thread obtains node numbers from a shared atomic counter and processes only the nodes whose numbers it obtained. Tickets are handed out in increasing order, and a thread requests a new ticket only after passing the node of its previous ticket. Hence the ticket a thread obtains is never smaller than its current position, and every node number below the total count is obtained by exactly one thread. So every subtree is processed exactly once, regardless of scheduling. Only the reported node counts depend on the number of threads, because the upper part of the tree is traversed by every thread.

## 6.2 Step 2: completion of the surviving cores

For each core  $C$  output by Step 1:

- (a) If  $U = \emptyset$ , then  $C$  satisfies (1) at every prime: at the primes of  $\mathcal{S}$  by the definition of  $U$ , and trivially at all other primes. So  $\Sigma(C) \in \mathbb{Z}$ . If  $C \neq \emptyset$ , then  $\Sigma(C) = 1$  by Corollary 2.3, and the program reports  $C$  as a solution. The empty core is not admissible, because  $V(K) \leq K/158 < 1$ .
- (b) If  $U \neq \emptyset$ , the program enumerates all  $(d, b)$  satisfying (S1) and (S2), discards those that violate (S3) or (S4), and processes the survivors. In (S4) the right-hand side is rounded up and  $1 - \Sigma(C)$  is rounded down, so no feasible shape is discarded. A surviving shape

**Table 3:** Recorded results of the two searches (`proof_run_output.txt`, `run47_output.txt`). Numbers marked <sup>†</sup> were obtained by the Python re-implementation of Step 2 (Section 8.2); all other numbers were printed by the C++ programs.

|                                                 | budget 46       | budget 47                                |
|-------------------------------------------------|-----------------|------------------------------------------|
| maximal number of leaves of a star (Lemma 5.2)  | 10              | 11                                       |
| search nodes (4 threads)                        | 139 830 520 211 | 1 795 181 713 099                        |
| cores kept by Step 1                            | 178             | 22 382                                   |
| of which $U = \emptyset$                        | 0               | 17                                       |
| of which $U \neq \emptyset$ and $ C  +  U  = K$ | 111             | 11 287                                   |
| shapes satisfying (S1)–(S3)                     | 3 777           | 1 107 702                                |
| shapes satisfying (S1)–(S4)                     | 61              | 8 049                                    |
| cores with at least one such shape              | 61              | 7 432                                    |
| with $b = 0$ / $b = 1$ / $b \geq 2$             | 61 / 0 / 0      | 7 981 <sup>†</sup> / 68 <sup>†</sup> / 0 |
| candidate stars (summed over surviving shapes)  | 0               | 125                                      |
| unresolved cores                                | 0               | 0                                        |
| probable (not proven) primes used               | –               | 0                                        |
| solutions found                                 | 0               | 23                                       |
| wall-clock time of the recorded run             | 19 min 40 s     | 5 h 48 min                               |

with  $b > 0$  is reported as *unresolved*. For a surviving shape with  $b = 0$ , the program lists all candidate stars with at most 10 leaves (Definition 5.3); it finds the centres by factoring  $n(A)$  completely. It then searches for an exact cover of the multiplicities by candidate stars with distinct centres that also satisfies the congruences (1) at every  $q \in W$ .

**Computational Claim 6.2** (Budget 46; Lean: `Search46`). *For every 46-admissible core  $C$ :*

- (i)  $U(C) \neq \emptyset$ ;
- (ii) every 46-feasible shape  $(d, b)$  for  $C$  has  $b = 0$ ;
- (iii) for every 46-feasible shape  $(d, 0)$  there is no candidate star with at most 10 leaves.

The recorded run of `semiprime_reciprocals.cpp` produced the output summarized in Table 3. The search visited  $1.4 \cdot 10^{11}$  nodes and kept 178 cores; none of them has  $U = \emptyset$ . Step 2 examined 3777 shapes satisfying (S1)–(S3). Only 61 of them also satisfy (S4), all with  $b = 0$ , and for none of these 61 shapes does a candidate star exist. Nothing was reported as unresolved and no solution was found. This is precisely Computational Claim 6.2; see Section 7.1 for the correspondence in detail. The run was repeated for this paper with identical results (Section 8.3).

*Proof of Theorem A, assuming Computational Claim 6.2.* Let  $T \subset \mathcal{P}$  be nonempty with  $|T| \leq 46$  and  $\Sigma(T) \in \mathbb{Z}$ . By Corollary 2.3,  $\Sigma(T) = 1$ , so  $T$  satisfies (2) with  $K = 46$ . Its core  $C$  is 46-admissible (Lemmas 4.3 and 4.4), so  $U \neq \emptyset$  by Computational Claim 6.2(i). The shape  $(d, b)$  of  $T$  is 46-feasible (Lemma 4.8), so  $b = 0$  by Computational Claim 6.2(ii). Choose  $q \in U$ . By Lemma 4.3,  $q$  has a large neighbour  $P$ . Since  $b = 0$ , all neighbours of  $P$  lie in  $\mathcal{S}$ . By Lemmas 5.1 and 5.2,  $(P, N(P))$  satisfies  $2 \leq |N(P)| \leq 10$  and  $P \mid n(N(P))$ , and  $N(P) \subseteq W$ . Moreover, if  $d_{q'} = 1$  for some  $q' \in N(P)$ , then  $P$  is the only large neighbour of  $q'$ , so  $P^{-1} \equiv -\rho(q') \pmod{q'}$  by (1). Thus  $(P, N(P))$  is a candidate star for  $(C, d)$ , contradicting Computational Claim 6.2(iii). Hence  $|T| \geq 47$ . Sharpness follows from Theorem B, or directly from any of the sets in Appendix A, whose validity is verified exactly and unconditionally (Section 8.1).  $\square$

## 7 Classification of the 47-term representations

The program `semiprime47.cpp` runs the same Step 1 with  $K = 47$ . Its Step 2 enumerates *all* completions instead of stopping at the first one:

- cores with  $U = \emptyset$  are recorded as solutions;
- for a surviving shape with  $b = 0$ , all exact covers by candidate stars with at most 11 leaves and distinct centres are enumerated, and a cover is kept if its value equals  $1 - \Sigma(C)$  *exactly*. The value is computed as an integer multiple of  $1/D$ , where  $D = \prod_{p \in \mathcal{S}} p < 2^{128}$  is a common denominator of all core and star values;
- for a surviving shape with  $b = 1$ , the program enumerates the following, as in [Section 5.2](#):
  - all unordered pairs  $\{A_1, A_2\}$  of nonempty subsets of  $W$  with  $[q \in A_1] + [q \in A_2] \leq d_q$ ;
  - all covers of the remaining multiplicities by candidate stars;
  - the integer  $t$ , if it exists;
  - all positive divisors  $X$  of  $N$ , obtained from the complete factorization of  $t + a_1 a_2$  and of  $b_1 b_2$ .

It keeps the pairs for which  $P_1 \neq P_2$  are primes larger than 73 and distinct from the star centres;

- a surviving shape with  $b \geq 2$  is reported as unresolved.

Every recorded set is then verified exactly: it has 47 elements, they are distinct squarefree semiprimes, and their reciprocal sum is 1 ([Section 8.1](#)).

**Computational Claim 7.1** (Budget 47; Lean: `Search47`). *Let  $\mathcal{T}_{23}$  be the family of the 23 sets listed in [Appendix A](#).*

- (i) *Every 47-admissible core  $C \neq \emptyset$  with  $U(C) = \emptyset$  belongs to  $\mathcal{T}_{23}$ .*
- (ii) *For every 47-admissible core with  $U \neq \emptyset$ , every 47-feasible shape has  $b \leq 1$ .*
- (iii) *( $b = 0$ ) Let  $C$  be 47-admissible with  $U \neq \emptyset$ , let  $(d, 0)$  be a 47-feasible shape, and let  $\{(P, A_P)\}_{P \in \mathcal{C}}$  be a family of candidate stars with at most 11 leaves and distinct centres such that each  $q \in \mathcal{S}$  is a leaf of exactly  $d_q$  of them and  $\Sigma(C) + \sum_P \sum_{q \in A_P} 1/(qP) = 1$ . Then  $C \cup \{qP : P \in \mathcal{C}, q \in A_P\} \in \mathcal{T}_{23}$ .*
- (iv) *( $b = 1$ ) Let  $C$  and  $(d, 1)$  be as in (iii), except that  $b = 1$ . Suppose we are given:*
  - *nonempty  $A_1, A_2 \subseteq \mathcal{S}$ ;*
  - *a family of candidate stars as in (iii) covering the multiplicities  $d_q - [q \in A_1] - [q \in A_2]$ , with total value  $\Sigma_F < 1 - \Sigma(C)$ ;*
  - *the positive integer  $t = (1 - \Sigma(C) - \Sigma_F) b_1 b_2$ ;*
  - *a positive divisor  $X$  of  $N$  such that  $P_1 = (X + a_1 b_2)/t$  and  $P_2 = (N/X + a_2 b_1)/t$  are distinct primes larger than 73 and are not centres of the stars.*

*Then the resulting set, consisting of  $C$ , the star elements,  $\{qP_1\}_{q \in A_1}$ ,  $\{qP_2\}_{q \in A_2}$  and  $P_1 P_2$ , belongs to  $\mathcal{T}_{23}$ .*

The recorded run ([Table 3](#)) kept 22 382 cores. Seventeen of them have  $U = \emptyset$ , all with  $|C| = 47$ , and these are the 17 solutions with all primes at most 73. Of the remaining 22 365 cores, 11 287 have  $|C| + |U| = 47$ . The printed count “ $|C| + |U| = 47$ : 11 287” in `run47_output.txt`

excludes the 17 cores with  $U = \emptyset$ . Including them, the dump of all kept cores contains 11 304 cores with  $|C| + |U| = 47$ . Step 2 examined 1 107 702 shapes. Of these, 8 049 shapes in 7 432 cores satisfy (S1)–(S4): 7 981 with  $b = 0$ , 68 with  $b = 1$ , and none with  $b \geq 2$ . For the 68 shapes with  $b = 1$  there are 7 896 admissible attachment pairs;  $t$  is an integer in 957 cases, and none of them yields a solution. The surviving shapes with  $b = 0$  have 125 candidate stars in total. They produce exactly the 6 solutions of Table 2, each consisting of a core with 44 elements and a single star with three leaves. No shape was unresolved. Every number declared prime is below  $2^{64}$ , where the primality test is deterministic, and “composite” verdicts of the Miller–Rabin test are always correct. So all primality decisions are rigorous (Section 10).

*Proof of Theorem B, assuming Computational Claim 7.1.* By Section 8.1, each of the 23 listed sets has 47 elements and reciprocal sum 1, and the sets are pairwise distinct. Conversely, let  $|T| = 47$  and  $\Sigma(T) = 1$ . Then (2) holds with  $K = 47$ , and the core  $C$  of  $T$  is 47-admissible.

- If  $U = \emptyset$ , then  $G = \emptyset$ . Indeed,  $C$  satisfies (1) everywhere, so  $\Sigma(C) \in \mathbb{Z}$ . Since  $V(47) < 1$ , the core  $C$  is nonempty, so  $\Sigma(C) = 1$  and  $\Sigma(G) = 0$ . Hence  $T = C \in \mathcal{T}_{23}$  by Computational Claim 7.1(i).
- If  $U \neq \emptyset$ , the shape of  $T$  is feasible, so  $b \leq 1$  by Computational Claim 7.1(ii).
  - If  $b = 0$ , the stars of  $T$  form a family as in Computational Claim 7.1(iii) (Section 5), so  $T \in \mathcal{T}_{23}$ .
  - If  $b = 1$ , Proposition 5.5 shows that the data  $(A_1, A_2, \text{stars}, t, X)$  of  $T$  are as in Computational Claim 7.1(iv), with  $X = tP_1 - a_1b_2 > 0$ . Hence  $T \in \mathcal{T}_{23}$ .

The statement about integral sums follows from Corollary 2.3. Parts (i) and (ii) of Theorem B are read off from the list; the numbers in Table 2 were recomputed exactly.  $\square$

**Structure of the 23 solutions.** All 23 solutions contain the 23 numbers 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 93, 95, 133, 155. Each solution involves between 14 and 16 distinct primes, and the union of the 23 solutions has 122 elements. The largest prime in the 17 solutions of type (i) is 71. Table 4 lists, for each solution, its largest prime, its core size, and its elements outside the common part.

## 7.1 Correspondence between the claims and the programs

Computational Claims 6.2 and 7.1 are stated with exact rational quantities, and the programs implement them as follows.

- **Step 1.** By Lemma 6.1, every admissible core is output by the search.
- **Shapes.** The shape enumeration generates every  $(d, b)$  satisfying (S1) and (S2). Since (S4) is tested with a rounded-up right-hand side, every feasible shape passes.
- **Candidate stars.** For every  $A \subseteq W$  with  $2 \leq |A| \leq m$ , the program removes all prime factors at most 73 from  $n(A)$  and factors the remainder completely. Each factor is certified prime by a deterministic test, and the product of the factors equals the remainder by construction. So every candidate star of Definition 5.3 is found. The residue condition for  $d_q = 1$  is tested exactly.
- **Covers.** Covers are enumerated over index-increasing sequences of candidates with distinct centres, so every family of Computational Claim 7.1(iii) is produced exactly once.

- **Case  $b = 1$ .** The program uses unordered pairs  $\{A_1, A_2\}$ , whereas [Computational Claim 7.1\(iv\)](#) quantifies over ordered pairs. Swapping  $(A_1, P_1) \leftrightarrow (A_2, P_2)$  replaces  $X$  by  $N/X$  and yields the same set.
- **Magnitudes.** All intermediate integers in the  $b = 1$  branch are stored in 128-bit words without explicit overflow checks. The Python recount ([Section 8.2](#)) shows that in the actual run the largest of them,  $N$ , has at most 59 bits. Moreover,  $b_1 b_2$ ,  $a_1 b_2$  and  $a_2 b_1$  have at most 31 bits and  $t + a_1 a_2$  at most 28 bits. So no overflow occurred.

This correspondence is an informal argument about the source code. It is *not* machine-checked; see [Section 9](#).

## 8 Exact and independent verification

We distinguish four layers of evidence: (1) the optimized C++ searches; (2) an independent Python re-implementation of Step 2; (3) exact verification of the 23 solutions; (4) the Lean formalization ([Section 9](#)). We also report checks performed specifically for this paper.

### 8.1 Exact verification of the 23 solutions

Each of the 23 sets was verified in four independent ways.

- `semiprime47.cpp` factors every element by trial division, certifies both factors with a deterministic primality test, and computes the reciprocal sum as a big-integer fraction over the product of all primes involved.
- `complete47.py` checks the sum with Python’s `fractions.Fraction`.
- In Lean, `Sol23_valid` proves that each set consists of 47 squarefree semiprimes, with explicit factorizations and primality proved by `norm_num`, and has reciprocal sum exactly 1. `Sol23_card` proves that the 23 sets are pairwise distinct.
- For this paper, the script `paper/verification/verify_paper_facts.py` rechecks everything with exact integer arithmetic and trial division. It checks distinctness of the elements and of the sets, the squarefree-semiprime property,  $\Sigma(T) = 1$ , and the congruences (1) at every vertex. It also recomputes all numbers quoted in [Sections 2](#) and [5](#) and generates all tables of this paper from the data.

### 8.2 Independent implementation of Step 2

The Python program `complete47.py` is a separate implementation of Step 2. It uses exact rationals (`fractions.Fraction`) and unbounded integers throughout, its own primality test and factorization (trial division followed by Pollard’s rho method [10]), and its own cover and divisor enumeration. It imposes no bound on the number of leaves of a candidate star, so it examines a superset of the candidates. It also recomputes  $U$  from each core and compares it with the value in the dump. Run on the file `cores47.txt.gz` of all 22 382 cores kept by Step 1 with  $K = 47$ , it finds the same 23 solutions and no unresolved core (`run47_python_crosscheck.txt`). For this paper we reran it with additional counters (`paper/verification/analyze47.py`, identical logic). It reproduces every count in the budget-47 column of [Table 3](#): 1 107 702 shapes, of which 8 049 survive in 7 432 cores, and 125 candidate stars (with or without the bound of 11 leaves). It also gives the split by  $b$  and the magnitudes quoted in [Section 7.1](#). The run takes about half a minute.

*Limitations.* The Python program checks Step 2 only. It consumes the output of the C++ Step 1 and cannot detect a core that Step 1 failed to output. It was written within the same project and implements the same mathematical reduction (Sections 4 and 5), so it is an independent *implementation*, not an independent *argument*.

### 8.3 Re-execution and the budget-46 cross-check

**Re-execution.** For this paper the budget-46 program `semiprime_reciprocals.cpp` was compiled and executed again with 4 threads (GCC 13.3.0; a cloud container with 4 virtual CPUs, Intel Xeon 2.10 GHz). The output is identical to the recorded `proof_run_output.txt`, including the node count 139 830 520 211, except for the elapsed time (1531 s, with other jobs running concurrently); log: `paper/verification/rerun/`.

**Budget-46 cores and a second Step 2.** The engine of `semiprime47.cpp`, included unmodified in a small driver, was run with budget 46; it writes every core kept by Step 1 to a file. It visited 139 830 520 211 nodes, kept 178 cores, examined 3 777 shapes of which 61 survive, found 0 candidate stars, 0 unresolved cores and 0 solutions, in agreement with Table 3. The Python Step 2 (`analyze47.py`, the logic of `complete47.py`) applied to these 178 cores with budget 46 likewise finds 3 777 shapes satisfying (S1)–(S3), 61 satisfying (S1)–(S4) (all with  $b = 0$ ), 0 candidate stars (also without the bound of 10 leaves), 0 unresolved shapes and 0 solutions. This archives the Python check of the budget-46 Step 2 mentioned in the repository documentation. Since the Step 1 code of both programs is the same, this is a second *execution* of Step 1, not an independent implementation.

### 8.4 Validation of Step 1 by brute force on reduced instances

Step 1 is the only part of the computation that has no second full-scale implementation. To test its code, we ran the engine of `semiprime47.cpp` on reduced instances, with  $\mathcal{S}$  replaced by the primes up to 13, 17 and 19 and with various budgets. We compared its output with a brute-force enumeration of *all*  $2^{15}$ ,  $2^{21}$  and  $2^{28}$  cores, respectively. For this test only, a copy of the program with Step 2 disabled was used; the Step 1 code is unchanged. For every core the brute force computes two sets, using the same fixed-point rounding as the program:

- SEG: the cores satisfying (P1),  $\Sigma(C) \leq 1$  exactly, and the rounded test (4) for *every* upper segment of  $U$  (Definition 4.5);
- FIN: the cores satisfying (P1), (P2) and the final test (4), i.e. only the case  $U' = U$ .

Lemma 6.1 and the form of the leaf test predict  $\text{SEG} \subseteq D \subseteq \text{FIN}$  for the set  $D$  of cores output by the search. This was confirmed in all 48 configurations tested: 12 pairs of cutoff and budget, the default table size of 8 low primes and a reduced size of 3 (which exercises the explicit recursion), and 1 and 3 threads. The search output never contained a duplicate. A negative control, which compared a budget-45 search with the reference sets for budget 50, was correctly reported as a failure. Representative sizes are in Table 5. They also show that FIN can be much larger than SEG, which quantifies Remark 4.6.

These tests exercise the same code path as the real search, including the table-driven enumeration, the residue bookkeeping, the  $U$ -accounting, all pruning tests and the parallel work distribution, on instances small enough to enumerate completely. They do not replace an independent full-scale implementation of Step 1.

## 8.5 Self-tests

The recorded self-tests (`selftest_output.txt`) provide further consistency checks.

- The engine, run without unsatisfied primes and with budget 47, finds exactly 17 solutions. These are the 17 solutions of [Theorem B\(i\)](#).
- The completion routine reconstructs a planted solution with the cutoff lowered to 53 and to 43. In the second case the shape has positive excess.
- The full pipeline with cutoff 53 and budget 47 finds 21 solutions, all among the 23. With cutoff 53 it also leaves 2902 unresolved cores; this is expected, because the parity and value tests are much weaker there, and it is not an error.

The engine of `semiprime47.cpp` also recovers a planted component with  $b = 1$  through the divisor equation (`run47_output.txt`).

## 8.6 What is not independently verified

The following are *not* independently verified:

- the full-scale Step 1 for either budget, which has a single implementation, validated only on reduced instances;
- the correctness of the compiler and the hardware;
- the informal correspondence of [Section 7.1](#) between [Computational Claims 6.2](#) and [7.1](#) and the source code.

The budget-46 Step 1 was, however, executed three times with identical results: in the recorded run, in a re-execution, and through the engine of the second program ([Section 8.3](#)).

## 9 Lean formalization

The directory `lean/` contains a Lean 4 [\[3\]](#) development based on Mathlib [\[13\]](#). It uses the toolchain `leanprover/lean4:v4.35.0-rc3` and the Mathlib commit

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and consists of nine files with about 2,700 lines and about 180 theorems and lemmas. It contains no `sorry`, `admit`, `axiom` or `native_decide`. The main theorems depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`, as reported by `#print axioms (lean/AxiomsCheck.lean)`. Auto-bound implicit variables are disabled. The development builds with `lake build`.

**Formally verified: the analytic reduction.** The following results are proved in Lean without any computational hypothesis:

- the local criterion ([Lemma 3.2](#); `integral_iff_localCond`);
- the exchange lemma, the list of the 47 smallest elements of  $\mathcal{P}$  together with the fact that it is complete, the inequalities of [Proposition 2.2](#), and [Corollary 2.3](#) (`recipSum_le_Aupto`, `sp_le_158`, `H37_lt_one`, `H38_gt_one`, `H47_lt_two`, `recipSum_eq_one_of_integral`, `card_ge_38`);
- [Lemma 4.3](#) (`card_Uset_le`), [Lemma 4.4](#) in the general form for every  $U' \subseteq U$  (`value_lower_subset`), and all parts of [Lemma 4.8](#) (`dcount_pos_of_mem_U`, `dcount_ne_one_of_not_mem_U`, `excess_le`, `parity_two`, `shape_value_bound`);

- the star lemmas and the star-size lemma (`star_dvd`, `star_card_ne_one`, `star_size`), the decomposition of  $G$  into stars and large-large edges (`bigPart_eq`, `recipSum_bigPart_eq`, `dcount_eq_mult`), and the finiteness of the candidate stars (`starCand_finite`);
- all parts of [Proposition 5.5](#) (`comp_value`, `comp_local_iff`, `comp_divisor_eq`, `comp_enum`, `comp_of_divisor`, `comp_pairs_finite`);
- the statement that the core and shape of every solution pass the tests of the search (`admissible_of_real`, `shapeOK_of_real`).

**Unconditional verification of the solutions.** `Sol23_valid` proves that each of the 23 sets consists of 47 squarefree semiprimes and has reciprocal sum 1, and `Sol23_card` proves that there are 23 distinct sets. The numerical facts are checked by the kernel through `decide` and `norm_num`. In particular, the existence part of [Theorem A](#) (sharpness) is fully formally verified.

**Conditional completeness.** The exhaustive searches are not reproduced in Lean. Instead, [Computational Claims 6.2](#) and [7.1](#) are stated as the propositions `Search46` and `Search47` (reproduced in [Appendix B](#)). These are ordinary definitions used as hypotheses, not axioms. Lean proves:

- `no_integral_sum_le_46`, which is [Theorem A](#) with `Search46` as a hypothesis, and its reformulation `card_ge_47_of_integral`;
- `solutions47_iff`, which is [Theorem B](#) with `Search47` as a hypothesis; `solutions47_integral_iff` states the version for integral sums, and `solutions47_ncard` states that the set of all 47-term representations of 1 has exactly 23 elements.

Consequently the complete proofs of [Theorems A](#) and [B](#) are formally verified *modulo* `Search46` and `Search47`, which are established only by the computations of [Sections 6](#) and [7](#). Two points are not verified in Lean:

- the fact that the C++ programs implement these propositions correctly ([Section 7.1](#));
- the fact that the recorded runs produced the stated output.

While the development was being written, the formalization exposed the two corrections of [Remarks 4.6](#) and [5.6](#). The Step 1 hypothesis is stated with the upper-segment condition of [Definition 4.5](#). With only the final value test, it would not describe what the search checks.

## 10 Computational reproducibility

**Arithmetic.** Neither C++ program uses a floating-point type (`float`, `double` or `long double`), nor any floating-point library function.

- Congruences are computed with machine integers modulo primes at most 73.
- Exact rational comparisons (the values  $H_k$ , the verification of solutions, the exact value tests of covers) use either a small arbitrary-precision natural-number type or 128-bit integers over the common denominator  $D = \prod_{p \in \mathcal{S}} p \approx 4.0 \cdot 10^{28}$ .
- Bounds on sums of reciprocals use the 96-bit fixed-point numbers up and lo of [Section 6.1](#). Upper bounds are always formed from up and lower bounds from lo. Every pruning decision compares an upper bound of a quantity that must be at least 1 with  $2^{96}$ , or a lower bound of a quantity that must be at most 1 with  $2^{96}$ .

Rounding can therefore only make a test *weaker*. It may keep a core or shape that exact arithmetic would discard, and this costs only time; it can never discard an admissible core or a feasible shape. All fixed-point sums are below  $3 \cdot 2^{96} < 2^{128}$ . The Python program uses exact rationals throughout.

**Primality and factorization.** Both programs decide primality of 64-bit integers with the Miller–Rabin test [11] to the bases  $2, 3, 5, \dots, 37$  (the first 12 primes). This test is deterministic for all  $n < \psi_{12} = 318\,665\,857\,834\,031\,151\,167\,461 \approx 3.2 \cdot 10^{23}$  [12], and in particular for all  $n < 2^{64}$ . The  $b = 1$  branch contains a probabilistic test for integers of at least  $2^{64}$ . It was never used: the run reports 0 probable primes. Factorizations use trial division and Pollard’s rho method [10]. The correctness of the result does not depend on the heuristics of the method, because every returned factor is certified prime and the factors multiply to the input by construction; only termination does, and the runs terminated. The inverses modulo  $p$  are computed as  $a^{p-2} \bmod p$ .

**Build and run.** The programs need a C++11 compiler with `unsigned __int128` (GCC or Clang):

```
g++ -O3 -march=native -pthread -o semiprime_reciprocals semiprime_reciprocals.cpp
./semiprime_reciprocals 4 # budget 46, 4 threads
g++ -O3 -march=native -pthread -o semiprime47 semiprime47.cpp
./semiprime47 4 --selftest --dump cores47.txt
gunzip -k cores47.txt.gz && python3 complete47.py cores47.txt 47
```

The recorded runs used 4 threads. The budget-46 search took 1179 s (19 min 40 s wall-clock), and the budget-47 run took 20 898 s for the search and 5 h 48 min wall-clock in total. The resident memory of the search is only a few megabytes; the virtual size of about 300 MB reported in the repository documentation consists mostly of thread stacks. The recorded outputs do not state the processor model. The re-execution of Section 8.3 used a cloud container with 4 virtual CPUs (Intel Xeon, 2.10 GHz) and GCC 13.3.0.

**Files.** The repository (*<repository URL to be inserted>*) contains the following files.

- Programs: `semiprime_reciprocals.cpp`, `semiprime47.cpp` and `complete47.py`.
- Recorded outputs: `proof_run_output.txt`, `selftest_output.txt`, `run47_output.txt` and `run47_python_crosscheck.txt`.
- Data: the core dump `cores47.txt.gz` and the list `solutions47.txt`.
- The Lean development `lean/`.
- The files of this paper in `paper/`, including the verification scripts, the brute-force test and the logs of the re-execution (`paper/verification/`).

## 11 Discussion

We have shown that at least 47 distinct squarefree semiprimes are needed to obtain an integral sum of reciprocals, and that exactly 23 sets of 47 such numbers have reciprocal sum 1. The minimality result confirms Watanabe’s conjecture [14].

The main mathematical point is the finite reduction. The local criterion (Lemma 3.2) turns integrality into a condition at each vertex of a graph. The core decomposition bounds the number of unsatisfied small primes. The star lemma and the component equation bound the large primes by divisibility, not by size. That large primes occur, for example 3779, shows

that a bound of this kind is unavoidable. A single large–large edge is handled by an explicit factorization identity. Configurations with two or more such edges are excluded here only by the computation: no feasible shape with  $b \geq 2$  exists for budget 47. For larger budgets this case would need further ideas.

**Limitations.** The proofs are computer-assisted.

- Their weakest point is Step 1, which has a single implementation. This implementation has been tested against brute force on reduced instances and executed three times for budget 46, but not reimplemented at full scale.
- Step 2 has been implemented twice, and the two implementations agree.
- The reduction and the 23 solutions are verified in Lean. The connection between the Lean hypotheses `Search46` and `Search47` and the C++ programs rests on the informal argument of [Section 7.1](#).

A fully formal proof would require the search to emit a certificate, such as the list of admissible cores with a refutation of each feasible shape, that can be checked in Lean. This is a natural direction for future work. An independent reimplement of Step 1, preferably by other authors and with a different enumeration strategy, would also strengthen the result considerably.

**Further questions.** The same method applies to other targets and budgets. Examples include the minimal number of terms for the integer 2, or for denominators with exactly three distinct prime factors (compare [\[1\]](#)). The cost of Step 1 grows quickly with the budget: from  $1.4 \cdot 10^{11}$  nodes for budget 46 to  $1.8 \cdot 10^{12}$  for budget 47. It would also be interesting to count the representations of 1 with 48 terms.

## 12 Disclosure of the use of generative AI

Generative AI tools, specifically Claude Code (Anthropic), an AI coding assistant based on large language models, were used substantially throughout this project, and not only for preparing the manuscript. Under the direction of the authors, the AI tool developed the mathematical reduction presented here, and it wrote the following:

- the C++ programs and the Python cross-check;
- the Lean formalization;
- the documentation files of the repository;
- the verification scripts written for this paper;
- a first draft of this manuscript.

The exploratory mathematical reasoning was also carried out with the tool. The two corrections of [Remarks 4.6](#) and [5.6](#) were found during the AI-assisted formalization and a later audit of the code.

All mathematical claims in this article were checked by the verification procedures described in [Sections 8](#) and [9](#):

- exact computation;
- a second implementation of Step 2;
- brute-force validation of Step 1 on reduced instances;

- a Lean 4 formalization of the analytic reduction and of the 23 solutions.

These checks were themselves designed and executed with the assistance of the same tool. The extent of the computer-assisted part and of the formally verified part, and the exact status of every result, are stated explicitly in [Figure 1](#) and [Sections 6 to 9](#). *[Editable by the authors: describe here any independent human checking of the code, the computations and the proofs that has been carried out.]* No AI system is an author of this work. The authors take full responsibility for its content.

## Acknowledgements

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## A The 23 representations of 1 with 47 terms

The 23 sets  $T \subset \mathcal{P}$  with  $|T| = 47$  and  $\Sigma(T) = 1$ , in the order of `solutions47.txt`. Solutions 1, 2, 3, 5, 11 and 12 contain a prime larger than 73 (Table 2).

1. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 74, 77, 82, 85, 87, 91, 93, 95, 111, 115, 119, 123, 133, 143, 145, 155, 203, 219, 221, 287, 299, 391, 481, 1299, 2117, 16021, 31609
2. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 74, 77, 82, 85, 87, 91, 93, 95, 111, 115, 119, 123, 133, 143, 155, 185, 203, 221, 287, 299, 319, 391, 407, 481, 933, 1555, 11507
3. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 74, 77, 82, 85, 87, 91, 93, 95, 111, 115, 119, 123, 133, 145, 155, 161, 187, 221, 253, 259, 287, 407, 629, 667, 1509, 6539, 14587
4. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 74, 77, 82, 85, 87, 91, 93, 95, 115, 119, 123, 129, 133, 143, 145, 155, 187, 213, 221, 287, 301, 559, 629, 781, 1247, 1591, 1633
5. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 74, 77, 82, 86, 87, 91, 93, 95, 106, 111, 115, 123, 133, 155, 159, 161, 185, 203, 215, 253, 265, 583, 667, 731, 109591, 139823, 154939
6. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 74, 77, 82, 86, 91, 93, 95, 111, 119, 122, 123, 133, 143, 145, 155, 161, 183, 187, 259, 287, 299, 319, 473, 481, 559, 671, 1591
7. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 74, 77, 85, 86, 87, 91, 93, 94, 95, 111, 115, 119, 133, 141, 142, 145, 155, 187, 213, 517, 629, 799, 1247, 1591, 1633, 2021, 2627
8. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 74, 77, 86, 87, 91, 93, 94, 95, 111, 115, 118, 119, 129, 133, 145, 155, 161, 185, 187, 329, 517, 667, 851, 1073, 1357, 1363, 2537
9. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 74, 82, 85, 86, 87, 91, 93, 95, 106, 111, 123, 133, 145, 155, 159, 185, 203, 215, 253, 265, 287, 319, 493, 583, 731, 851, 1073
10. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 77, 82, 85, 86, 87, 91, 93, 94, 95, 111, 115, 119, 133, 141, 142, 155, 187, 205, 213, 235, 493, 517, 1363, 1633, 1763, 1927, 2627
11. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 77, 82, 85, 86, 87, 91, 93, 94, 95, 115, 118, 119, 123, 129, 133, 141, 155, 203, 287, 319, 391, 799, 1345, 1357, 2537, 2959, 12643
12. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 77, 82, 85, 86, 87, 91, 93, 94, 95, 115, 118, 119, 123, 133, 143, 145, 155, 203, 221, 287, 391, 559, 1357, 2537, 12209, 18103, 19787
13. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 77, 82, 85, 86, 87, 91, 93, 94, 95, 115, 119, 123, 129, 133, 141, 142, 145, 155, 187, 517, 799, 1207, 1247, 1633, 1763, 2021, 2911
14. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 77, 82, 85, 86, 87, 91, 93, 94, 95, 118, 119, 123, 129, 133, 155, 161, 187, 213, 287, 329, 355, 493, 497, 517, 1357, 1363, 2537
15. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 77, 82, 85, 86, 87, 91, 93, 94, 95, 118, 119, 123, 133, 145, 155, 161, 187, 203, 215, 287, 329, 493, 517, 1247, 1357, 1363, 2537
16. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 77, 82, 85, 86, 87, 93, 94, 95, 115, 118, 119, 123, 129, 133, 141, 143, 145, 155, 287, 377, 391, 551, 799, 893, 1357, 1363, 2537
17. 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 77, 82, 85, 86, 87, 93, 94, 95, 115, 118, 119, 123, 129, 133, 143, 145, 155, 177, 287, 299, 391, 551, 611, 799, 893, 1711, 2537

- 18.** 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 77, 82, 85, 86, 91, 93, 94, 95, 115, 119, 123, 133, 141, 142, 143, 155, 203, 221, 235, 287, 299, 355, 377, 391, 559, 2021, 2059
- 19.** 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 77, 82, 85, 87, 91, 93, 94, 95, 111, 115, 118, 119, 123, 133, 142, 155, 187, 213, 287, 295, 329, 413, 493, 517, 1363, 1633, 2627
- 20.** 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 77, 82, 86, 87, 91, 93, 94, 95, 118, 119, 123, 129, 133, 145, 155, 161, 177, 187, 203, 287, 299, 329, 377, 517, 1363, 1711, 2537
- 21.** 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 77, 82, 86, 91, 93, 94, 95, 111, 115, 119, 123, 133, 141, 142, 143, 145, 155, 203, 213, 221, 235, 287, 493, 559, 1633, 2021, 2627
- 22.** 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 69, 77, 82, 87, 91, 93, 95, 106, 111, 115, 119, 122, 123, 133, 142, 155, 159, 183, 187, 203, 213, 265, 287, 319, 583, 671, 1633, 2627
- 23.** 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58, 62, 65, 74, 77, 82, 85, 87, 91, 93, 95, 111, 115, 119, 123, 129, 133, 143, 145, 155, 161, 221, 253, 259, 287, 299, 391, 473, 481, 1247, 1591

## B The computational hypotheses in Lean

The hypotheses of [Section 9](#), verbatim from `lean/SemiprimeEgypt/Main.lean`.

- `CoreSetC` says that every element of  $C$  lies in  $\mathcal{P}_S$ .
- `AdmissibleKC` is [Definition 4.5](#).
- `ShapeOKKCdb` is [Definition 4.7](#).
- `StarCandmCdPA` is [Definition 5.3](#).
- `mult`, `starEdges` and `starVal` are the multiplicities, elements and value of a star family.
- `compN` and `compP\(_1\)`, `compP\(_2\)` are  $N$ ,  $P_1$  and  $P_2$  of [Proposition 5.5](#).
- `Sol23` is the family of [Appendix A](#).

```
def Search46 : Prop :=
  ∀ C : Finset ℕ, CoreSet C → Admissible 46 C →
    (Uset 73 C).Nonempty ∧
    ∀ (d : ℕ → ℕ) (b : ℕ), ShapeOK 46 C d b → b = 0 ∧ ∀ P A, ¬ StarCand 10 C d P A

def Search47 : Prop :=
  (∀ C : Finset ℕ, CoreSet C → Admissible 47 C → Uset 73 C = ∅ → C.Nonempty → C ∈ Sol23) ∧
  (∀ C : Finset ℕ, CoreSet C → Admissible 47 C → (Uset 73 C).Nonempty →
    ∀ (d : ℕ → ℕ) (b : ℕ), ShapeOK 47 C d b → b ≤ 1) ∧
  (∀ (C : Finset ℕ) (d : ℕ → ℕ) (Ps : Finset ℕ) (A : ℕ → Finset ℕ),
    CoreSet C → Admissible 47 C → (Uset 73 C).Nonempty → ShapeOK 47 C d 0 →
    (∀ P ∈ Ps, StarCand 11 C d P (A P)) →
    (∀ q ∈ smallPrimes 73, mult Ps A q = d q) →
    recipSum C + starVal Ps A = 1 →
    C ∪ starEdges Ps A ∈ Sol23) ∧
  (∀ (C : Finset ℕ) (d : ℕ → ℕ) (Ps : Finset ℕ) (A : ℕ → Finset ℕ) (A1 A2 : Finset ℕ)
    (t X : ℕ),
    CoreSet C → Admissible 47 C → (Uset 73 C).Nonempty → ShapeOK 47 C d 1 →
    (∀ P ∈ Ps, StarCand 11 C d P (A P)) →
    A1.Nonempty → A2.Nonempty → A1 ⊆ smallPrimes 73 → A2 ⊆ smallPrimes 73 →
    (∀ q ∈ smallPrimes 73,
      mult Ps A q + (if q ∈ A1 then 1 else 0) + (if q ∈ A2 then 1 else 0) = d q) →
    recipSum C + starVal Ps A < 1 →
    0 < t →
    (t : ℚ) = (1 - recipSum C - starVal Ps A) * (∏ q ∈ A1, (q : ℚ)) * ∏ q ∈ A2, (q : ℚ) →
    0 < X → X | compN A1 A2 t → t | X + nA A1 * ∏ q ∈ A2, q →
    t | compN A1 A2 t / X + nA A2 * ∏ q ∈ A1, q →
```

```

compP1 A1 A2 t X ≠ compP2 A1 A2 t X →
73 < compP1 A1 A2 t X → 73 < compP2 A1 A2 t X →
(compP1 A1 A2 t X).Prime → (compP2 A1 A2 t X).Prime →
compP1 A1 A2 t X ∉ Ps → compP2 A1 A2 t X ∉ Ps →
C ∪ starEdges Ps A ∪ A1.image (· * compP1 A1 A2 t X) ∪
  A2.image (· * compP2 A1 A2 t X) ∪ {compP1 A1 A2 t X * compP2 A1 A2 t X} ∈ Sol23)

```

**Table 4:** The 23 solutions of [Theorem B](#). Each solution consists of the 23 common elements listed in the text together with the 24 elements in the last column. “Core” is the number of elements with both prime factors at most 73.

| no. | max. prime | core | $P > 73$ | remaining 24 elements                                                                                                    |
|-----|------------|------|----------|--------------------------------------------------------------------------------------------------------------------------|
| 1   | 433        | 44   | 433      | 69, 74, 77, 82, 85, 87, 91, 111, 115, 119, 123, 143, 145, 203, 219, 221, 287, 299, 391, 481, 1299, 2117, 16021, 31609    |
| 2   | 311        | 44   | 311      | 69, 74, 77, 82, 85, 87, 91, 111, 115, 119, 123, 143, 185, 203, 221, 287, 299, 319, 391, 407, 481, 933, 1555, 11507       |
| 3   | 503        | 44   | 503      | 69, 74, 77, 82, 85, 87, 91, 111, 115, 119, 123, 145, 161, 187, 221, 253, 259, 287, 407, 629, 667, 1509, 6539, 14587      |
| 4   | 71         | 47   | –        | 69, 74, 77, 82, 85, 87, 91, 115, 119, 123, 129, 143, 145, 187, 213, 221, 287, 301, 559, 629, 781, 1247, 1591, 1633       |
| 5   | 3779       | 44   | 3779     | 69, 74, 77, 82, 86, 87, 91, 106, 111, 115, 123, 159, 161, 185, 203, 215, 253, 265, 583, 667, 731, 109591, 139823, 154939 |
| 6   | 61         | 47   | –        | 69, 74, 77, 82, 86, 91, 111, 119, 122, 123, 143, 145, 161, 183, 187, 259, 287, 299, 319, 473, 481, 559, 671, 1591        |
| 7   | 71         | 47   | –        | 69, 74, 77, 85, 86, 87, 91, 94, 111, 115, 119, 141, 142, 145, 187, 213, 517, 629, 799, 1247, 1591, 1633, 2021, 2627      |
| 8   | 59         | 47   | –        | 69, 74, 77, 86, 87, 91, 94, 111, 115, 118, 119, 129, 145, 161, 185, 187, 329, 517, 667, 851, 1073, 1357, 1363, 2537      |
| 9   | 53         | 47   | –        | 69, 74, 82, 85, 86, 87, 91, 106, 111, 123, 145, 159, 185, 203, 215, 253, 265, 287, 319, 493, 583, 731, 851, 1073         |
| 10  | 71         | 47   | –        | 69, 77, 82, 85, 86, 87, 91, 94, 111, 115, 119, 141, 142, 187, 205, 213, 235, 493, 517, 1363, 1633, 1763, 1927, 2627      |
| 11  | 269        | 44   | 269      | 69, 77, 82, 85, 86, 87, 91, 94, 115, 118, 119, 123, 129, 141, 203, 287, 319, 391, 799, 1345, 1357, 2537, 2959, 12643     |
| 12  | 421        | 44   | 421      | 69, 77, 82, 85, 86, 87, 91, 94, 115, 118, 119, 123, 143, 145, 203, 221, 287, 391, 559, 1357, 2537, 12209, 18103, 19787   |
| 13  | 71         | 47   | –        | 69, 77, 82, 85, 86, 87, 91, 94, 115, 119, 123, 129, 141, 142, 145, 187, 517, 799, 1207, 1247, 1633, 1763, 2021, 2911     |
| 14  | 71         | 47   | –        | 69, 77, 82, 85, 86, 87, 91, 94, 118, 119, 123, 129, 161, 187, 213, 287, 329, 355, 493, 497, 517, 1357, 1363, 2537        |
| 15  | 59         | 47   | –        | 69, 77, 82, 85, 86, 87, 91, 94, 118, 119, 123, 145, 161, 187, 203, 215, 287, 329, 493, 517, 1247, 1357, 1363, 2537       |
| 16  | 59         | 47   | –        | 69, 77, 82, 85, 86, 87, 94, 115, 118, 119, 123, 129, 141, 143, 145, 287, 377, 391, 551, 799, 893, 1357, 1363, 2537       |
| 17  | 59         | 47   | –        | 69, 77, 82, 85, 86, 87, 94, 115, 118, 119, 123, 129, 143, 145, 177, 287, 299, 391, 551, 611, 799, 893, 1711, 2537        |
| 18  | 71         | 47   | –        | 69, 77, 82, 85, 86, 91, 94, 115, 119, 123, 141, 142, 143, 203, 221, 235, 287, 299, 355, 377, 391, 559, 2021, 2059        |
| 19  | 71         | 47   | –        | 69, 77, 82, 85, 87, 91, 94, 111, 115, 118, 119, 123, 142, 187, 213, 287, 295, 329, 413, 493, 517, 1363, 1633, 2627       |
| 20  | 59         | 47   | –        | 69, 77, 82, 86, 87, 91, 94, 118, 119, 123, 129, 145, 161, 177, 187, 203, 287, 299, 329, 377, 517, 1363, 1711, 2537       |
| 21  | 71         | 47   | –        | 69, 77, 82, 86, 91, 94, 111, 115, 119, 123, 141, 142, 143, 145, 203, 213, 221, 235, 287, 493, 559, 1633, 2021, 2627      |
| 22  | 71         | 47   | –        | 69, 77, 82, 87, 91, 106, 111, 115, 119, 122, 123, 142, 159, 183, 187, 203, 213, 265, 287, 319, 583, 671, 1633, 2627      |
| 23  | 43         | 47   | –        | 74, 77, 82, 85, 87, 91, 111, 115, 119, 123, 129, 143, 145, 161, 221, 253, 259, 287, 299, 391, 473, 481, 1247, 1591       |

**Table 5:** Brute-force validation of Step 1 on reduced instances ( $\mathcal{S}$  = primes up to  $s$ ).  $D$  is the output of the search, with the default table of the 8 smallest primes and with a table of 3 primes. The output did not depend on the number of threads. In all cases  $\text{SEG} \subseteq D \subseteq \text{FIN}$ .

| $s$ | edges | budget | SEG     | $D$  , table size |         | FIN       |
|-----|-------|--------|---------|-------------------|---------|-----------|
|     |       |        |         | 8                 | 3       |           |
| 13  | 15    | 40     | 13      | 13                | 13      | 140       |
| 13  | 15    | 45     | 156     | 161               | 161     | 512       |
| 13  | 15    | 50     | 539     | 586               | 586     | 1 284     |
| 13  | 15    | 60     | 2 083   | 2 390             | 2 390   | 3 872     |
| 13  | 15    | 80     | 7 260   | 8 492             | 8 492   | 10 774    |
| 17  | 21    | 40     | 32      | 32                | 32      | 484       |
| 17  | 21    | 45     | 2 388   | 2 422             | 2 422   | 10 001    |
| 17  | 21    | 50     | 13 111  | 13 669            | 13 669  | 36 845    |
| 17  | 21    | 60     | 72 019  | 79 084            | 79 083  | 139 032   |
| 19  | 28    | 40     | 20      | 20                | 20      | 2 535     |
| 19  | 28    | 45     | 38 109  | 38 109            | 38 109  | 175 804   |
| 19  | 28    | 50     | 475 652 | 478 084           | 478 084 | 1 146 681 |